

Example #06

Find the vector projection of a force  $F = 5i + 2j$  onto  $v = i - 3j$  and the scalar component of  $F$  in the direction of  $v$ .

Sol:-

vector Projection

$$F = \left( \frac{F \cdot v}{|v|^2} \right) v$$

$$\begin{aligned} F \cdot v &= (5 \times 1)i + (2 \times (-3))j \\ &= 5i + (-6j) \\ &= 5 - 6 \\ &= -1 \end{aligned}$$

$$\begin{aligned} |v| &= \sqrt{(v_1)^2 + (v_2)^2} \Rightarrow v_1 = 5 \quad v_2 = -6 \\ &= \sqrt{(5)^2 + (-6)^2} \\ &= \sqrt{25 + 36} \end{aligned}$$

$$\begin{aligned} |v| &= \sqrt{(v_1)^2 + (v_2)^2} \Rightarrow v_1 = 1 \quad v_2 = -3 \\ |v| &= \sqrt{(1)^2 + (-3)^2} \\ |v| &= \sqrt{1 + 9} \\ |v| &= \sqrt{10} \end{aligned}$$

Put into function

$$F = \left( \frac{F \cdot v}{|v|^2} \right) v \Rightarrow \left( \frac{-1}{(\sqrt{10})^2} \right) (i - 3j)$$

$$F = \left( \frac{-1}{10} \right) (i - 3j) \Rightarrow -\frac{1}{10}i + \frac{3}{10}j$$

The Scalar Component of  $F$  in the direction of  $v$  is

$$|F| \cos \theta = \frac{F \cdot v}{|v|} = \frac{-1}{\sqrt{10}}$$

# Ex# 12.3

Example # 07

If  $|F| = 40\text{ N}$ ,  $|D| = 3\text{ m}$  and  $\theta = 60^\circ$  the work done by  $F$  in acting from  $P$  to  $Q$  is

$$\text{work} = F \cdot D \Rightarrow \text{Displacement}$$

$$= (|F| \cos \theta) |D|$$

$$= (40 \cos 60) 3$$

$$(20) 3$$

$$60\text{ J}$$

## Problems

### Dot Product and Projections

01.  $v = 2i - 4j + \sqrt{5}k$ ,  $u = -2i + 4j - \sqrt{5}k$

Sol.

a.  $v \cdot u$ ,  $|v|$ ,  $|u|$

$$\begin{aligned} v \cdot u &= (v_1 \times u_1) + (v_2 \times u_2) + (v_3 \times u_3) \\ &= (2 \times (-2)) + (-4 \times 4) + (\sqrt{5} \times -\sqrt{5}) \\ &= (-4) + (-16) + \cancel{(\sqrt{5} \times \sqrt{5})} \\ &= -20 - 5 \\ &= -25 \end{aligned}$$

$$|v| = \sqrt{(v_1)^2 + (v_2)^2 + (v_3)^2} \Rightarrow \sqrt{(2)^2 + (-4)^2 + (\sqrt{5})^2}$$

$$\sqrt{4 + 16 + 5} \Rightarrow \sqrt{25} = 5$$

$$|u| = \sqrt{(u_1)^2 + (u_2)^2 + (u_3)^2} \Rightarrow \sqrt{(-2)^2 + (4)^2 + (-\sqrt{5})^2}$$

$$\sqrt{4 + 16 + 5} \Rightarrow \sqrt{25} = 5$$

b = The Cosine of the angle between  $v$  and  $u$

$$\cos(\theta) = \frac{v \cdot u}{|v||u|} \Rightarrow \frac{-25}{5 \times 5} \Rightarrow \frac{-25}{25} = -1$$

c = Scalar Component of  $u$  in the direction of

$$v = (v \cdot u) / |v| \Rightarrow \frac{-25}{5} = -5$$

d = the vector Project

$$\text{Proj}_v u = \left( \frac{u \cdot v}{|v|^2} \right) v$$

$$u = \left( \frac{-25}{15^2} \right) (2i - 4j + \sqrt{5}k)$$

$$= \left( \frac{-25}{25} \right) 2i - 4j + \sqrt{5}k$$

$$= -2i + 4j - \sqrt{5}k$$

$$02) \quad v = \frac{3}{5}i + \frac{4}{5}k \quad u = 5i + 12j$$

Sol:-

$$v \cdot u = \frac{3}{5}i \times 5i$$

$$\left(\frac{3 \cdot 5}{5}\right)i \Rightarrow 3$$

$$\begin{aligned} |v| &= \sqrt{(v_1)^2 + (v_2)^2} \\ &= \sqrt{(3/5)^2 + (4/5)^2} \\ &= \sqrt{1} = 1 \end{aligned}$$

$$\begin{aligned} |u| &= \sqrt{(u_1)^2 + (u_2)^2} \\ &= \sqrt{(5)^2 + (12)^2} \\ &= \sqrt{25 + 144} \\ &= \sqrt{169} = 13 \end{aligned}$$

$$b) \quad \cos(\theta) = \frac{(v \cdot u)}{|v||u|} = \frac{3}{1 \times 13} \Rightarrow \frac{3}{13}$$

$$c) \quad v = \frac{(v \cdot u)}{|v|} = \frac{3}{1} = 3$$

$$\begin{aligned} d) \quad u &= \left( \frac{u \cdot v}{|v|^2} \right) v \\ &= \frac{3}{1} \left( \frac{3}{(1)^2} \right) \left( \frac{3}{5}i + \frac{4}{5}k \right) \end{aligned}$$

$$\frac{9}{5}i + \frac{12}{5}k$$

03)  $v = 10i + 11j - 2k$  ,  $u = 3j + 4k$

Sol:-

~~$v \cdot u = 11j$~~

$$\begin{aligned} v \cdot u &= (11 \times 3)j + (4 \times (-2))k \\ &= 33j + (-8k) \\ &= 33 - 8 \\ &= 25 \end{aligned}$$

$$\begin{aligned} |v| &= \sqrt{(v_1)^2 + (v_2)^2 + (v_3)^2} \\ &= \sqrt{(10)^2 + (11)^2 + (-2)^2} \\ &= \sqrt{100 + 121 + 4} \\ &= \sqrt{225} = 15 \end{aligned}$$

$$\begin{aligned} |u| &= \sqrt{(u_1)^2 + (u_2)^2 + (u_3)^2} \\ &= \sqrt{(3)^2 + (4)^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} = 5 \end{aligned}$$

$$b) \cos(\theta) = \frac{v \cdot u}{|v| |u|} = \frac{25}{15 \times 5} = \frac{25}{75} = \frac{1}{3}$$

$$c) v = \frac{(v \cdot u)}{|v|} = \frac{25}{15 \times 3} = \frac{5}{3}$$

$$\begin{aligned} d) u &= \left( \frac{u \cdot v}{|v|^2} \right) v \\ &= \left( \frac{25}{(15)^2} \right) 10i + 11j - 2k \Rightarrow \left( \frac{25}{225} \right) = \frac{1}{9} \end{aligned}$$

$$\frac{1}{9} (10i + 11j - 2k)$$

$$Q4) \quad v = 2i + 10j - 11k \quad u = 2i + 2j + k$$

Sol:-

$$\begin{aligned} v \cdot u &= (v_1 \times u_1) + (v_2 \times u_2) + (v_3 \times u_3) \\ &= (2 \times 2) + (10 \times 2) + (-11 \times 1) \\ &= 4 + 20 - 11 \\ &= 13 \end{aligned}$$

$$\begin{aligned} |v| &= \sqrt{(v_1)^2 + (v_2)^2 + (v_3)^2} \\ &= \sqrt{(2)^2 + (10)^2 + (-11)^2} \\ &= \sqrt{4 + 100 + 121} \\ &= \sqrt{225} \\ &= 15 \end{aligned}$$

$$\begin{aligned} |u| &= \sqrt{(u_1)^2 + (u_2)^2 + (u_3)^2} \\ &= \sqrt{(2)^2 + (2)^2 + (1)^2} \\ &= \sqrt{4 + 4 + 1} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

$$B) \quad \frac{u \cdot v}{|v||u|} = \frac{13}{15 \times 3} = \frac{13}{45}$$

$$C) \quad \frac{u \cdot v}{|v|} = \frac{13}{15}$$

$$d) \quad \left( \frac{u \cdot v}{|v|^2} \right) v \Rightarrow \left( \frac{13}{(15)^2} \right) (2i + 10j - 11k)$$

$$\frac{13}{225} (2i + 10j - 11k)$$

Angle Between vectors

09)  $u = 2i + j$      $v = i + 2j - k$

Sol:-

$$\theta = \cos^{-1} \left( \frac{u \cdot v}{|u||v|} \right)$$

$$u \cdot v = (2 \times 1) + (1 \times 2) \\ = 2 + 2$$

$$|u| = \sqrt{(u_1)^2 + (u_2)^2 + (u_3)^2} \\ = \sqrt{(1)^2 + (2)^2 + (-1)^2} \\ = \sqrt{1 + 4 + 1} \\ = \sqrt{6}$$

$$|v| = \sqrt{(v_1)^2 + (v_2)^2} \\ = \sqrt{(2)^2 + (1)^2} \\ = \sqrt{4 + 1} \\ = \sqrt{5}$$

$$\theta = \cos^{-1} \left( \frac{u \cdot v}{|u||v|} \right) \\ = \cos^{-1} \left( \frac{4}{\sqrt{5} \sqrt{6}} \right) \Rightarrow \left( \frac{4}{\sqrt{5 \times 6}} \right)$$

$$\cos^{-1} \left( \frac{4}{\sqrt{30}} \right)$$

$$10) \quad u = 2i - 2j + k, \quad v = 3i + 4k$$

Sol:-

$$\begin{aligned} u \cdot v &= (u_1 \times v_1) + (u_2 \times v_2) \\ &= (2 \times 3) + (1 \times 4) \\ &= 6 + 4 \\ &= 10 \end{aligned}$$

$$\begin{aligned} |v| &= \sqrt{(v_1)^2 + (v_2)^2} \\ &= \sqrt{(3)^2 + (4)^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

$$\begin{aligned} |u| &= \sqrt{(u_1)^2 + (u_2)^2 + (u_3)^2} \\ &= \sqrt{(2)^2 + (-2)^2 + (1)^2} \\ &= \sqrt{4 + 4 + 1} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

$$\theta = \cos^{-1} \left( \frac{u \cdot v}{|u| |v|} \right)$$

$$= \cos^{-1} \left( \frac{10}{5 \times 3} \right) = \frac{10}{15}$$

$$\cos^{-1} \left( \frac{2}{3} \right)$$

13) Triangle find the measures of the angles of the triangle whose vertices are  $A = (-1, 0)$ ,  $B = (2, 1)$ ,  $C = (1, -2)$   
Sol:-

$$\begin{aligned} \vec{AB} &= (x_2 - x_1, y_2 - y_1) & \vec{AC} &= (x_2 - x_1, y_2 - y_1) \\ \vec{AB} &= (2 - (-1), 1 - 0) & &= (1 - (-1), (-2) - 0) \\ &= (3, 1) & &= (2, -2) \end{aligned}$$

$$\begin{aligned} \vec{BC} &= (x_2 - x_1, y_2 - y_1) & \vec{BA} &= (x_2 - x_1, y_2 - y_1) \\ &= (1 - 2, -2 - 1) & &= (-1 - 2, 0 - 1) \\ &= (-1, -3) & &= (-3, -1) \end{aligned}$$

$$\begin{aligned} \vec{CB} &= (x_2 - x_1, y_2 - y_1) & \vec{CA} &= (x_2 - x_1, y_2 - y_1) \\ &= (2 - 1, 1 - (-2)) & &= (-1 - 1, 0 - (-2)) \\ &= (1, 3) & &= (-2, 2) \end{aligned}$$

$$\text{Angle at A} = \cos^{-1} \left( \frac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}| |\vec{AC}|} \right) = \frac{3(2) + 1(-2)}{(\sqrt{10}) \sqrt{8}}$$

$$\cos^{-1} \left( \frac{6 - 2}{(\sqrt{10})(\sqrt{8})} \right) = \frac{4}{(\sqrt{10})(\sqrt{8})}$$

$$\text{Angle at B} = \cos^{-1} \left( \frac{\vec{BC} \cdot \vec{BA}}{|\vec{BC}| |\vec{BA}|} \right) = \frac{3 + 3}{(\sqrt{10})(\sqrt{10})} \Rightarrow \frac{6}{10} = \frac{3}{5}$$

$$\text{Angle at C} = \cos^{-1} \left( \frac{\vec{CB} \cdot \vec{CA}}{|\vec{CB}| |\vec{CA}|} \right) = \frac{-2 + 6}{(\sqrt{10})(\sqrt{8})} = \frac{4}{(\sqrt{10})(\sqrt{8})}$$

14) Rectangle find the measures of the angles between the diagonals of the rectangle whose vertices are  $A = (1, 0)$   $B = (0, 3)$   $C = (3, 4)$   $D = (4, 1)$

Sol:-

$$\begin{aligned}\vec{AC} &= (x_2 - x_1, y_2 - y_1) \\ &= (3 - 1, 4 - 0) \\ &= (2, 4)\end{aligned}$$

$$\begin{aligned}\vec{BD} &= (x_2 - x_1, y_2 - y_1) \\ &= (4 - 0, 1 - 3) \\ &= (4, -2)\end{aligned}$$

$$\begin{aligned}\vec{AC} \cdot \vec{BD} &= 2(4) + 4(-2) \\ &= 8 - 8 \\ &= 0\end{aligned}$$